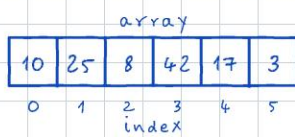


Mathwriter Diagram System - Comprehensive Review

This document demonstrates the complete diagram generation system.

Every diagram below is drawn using the DRAW engine from low-level primitives.

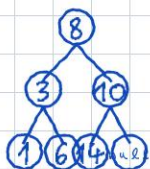
1. Array Indexing



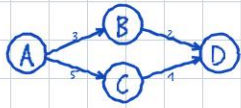
2. Singly Linked List



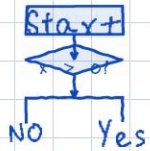
3. Binary Search Tree



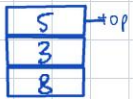
4. Weighted Directed Graph



5. Flowchart (if-else decision)



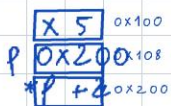
6. Stack (push/pop)



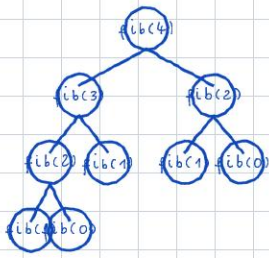
7. DP Table with Cell Highlighting

0	1	2	3
1	1	2	3
2	2	3	+

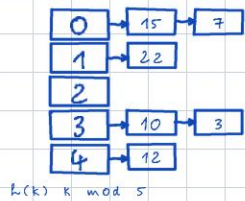
8. Memory Layout with Pointers



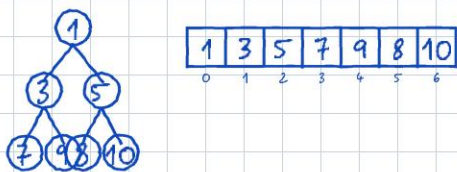
9. Recursion Tree (Fibonacci)



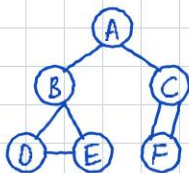
10. Hash Table with Chaining



11. Min-Heap (Tree + Array)

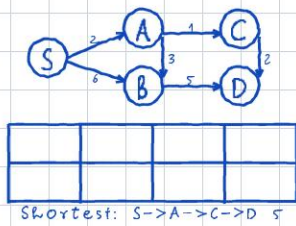


12. BFS Traversal

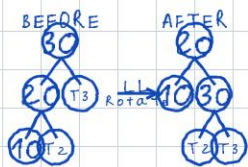


Queue: (A) → (B,C) → (C,D,E) → (D,E,F) → (E,F) → (F) → ()
Order: A, B, C, D, E, F

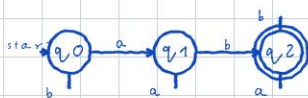
13. Dijkstra Shortest Path



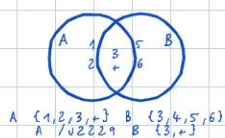
14. AVL Tree Rotation (LL Case)



15. DFA State Machine



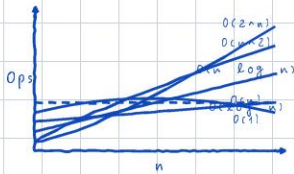
16. Venn Diagram



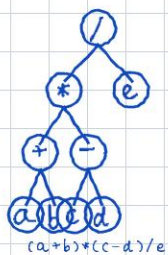
17. Number Line



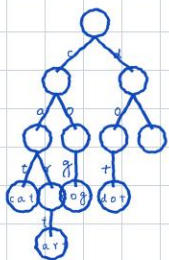
18 Big-O Complexity Comparison



19. Expression Tree



20. Trie (Prefix Tree)



System Capabilities Summary

The DRAW engine supports 16 primitives with 5 line styles:

| Primitive | Purpose |

|-----|-----|

| LINE | Straight lines, axes, connectors |

| CURVE | Quadratic bezier curves |

| CUBIC | cubic bezier curves |

| RECT | Boxes, memory cells, nodes |

| CIRCLE | Tree/graph nodes, sets |

| ELLIPSE | Oval shapes |

| ARROW | Directed edges, pointers, flow |

| PATH | Multi-point open paths |

| POLYGON | Diamonds, triangles, closed shapes |

| ARC | Circular arcs, pie slices |

| GRID | Tables, matrices, arrays |

| TEXT | Labels, values, annotations |

| DOT | Points, markers, filled nodes |

| BRACKET | Parentheses, grouping |

| BRACE | Curly braces, Set notation |

| HIGHLIGHT | Emphasis, cell highlighting |

Line styles: wobbly (default), smooth, dashed, rough, thick.

The engine can draw ANY computer science diagram - trees, graphs, arrays, linked lists, stacks, queues, heaps, hash tables, flowcharts, state machines, memory layouts, number lines, Venn diagrams, coordinate planes, function plots, tries, expression trees, DP tables, and more. All in hand-drawn Apple Pencil

blue ink style.